Unusual Does Not Mean Impossible

Topic 4.11 · TODAY'S PROBLEM

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: "Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies."

Analyst: "A result can be unusual without being impossible; compare 58 with the model's mean and standard deviation."

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE (5 min, on your sheet)

Whose statement is more defensible, the coach's or the analyst's? Name one quantity you would check.

DOK 1 **(a)** State n and p for X and write the formulas for the binomial mean and standard deviation.

DOK 2 **(b)** Without calculating yet, interpret the model's mean and standard deviation in the context of many sets of 100 free throws.

DOK 3 **(c)** Compute μ_X, σ_X, and the z-score of 58. Check np and $n(1-p)$ to decide whether a normal approximation is reasonable. Is 58 unusual? Adjudicate the two statements, distinguish "unusual" from "impossible," and explain what the coach's claim could mislead a reader into believing.

Video + follow-along: `u4_lesson10-12_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.